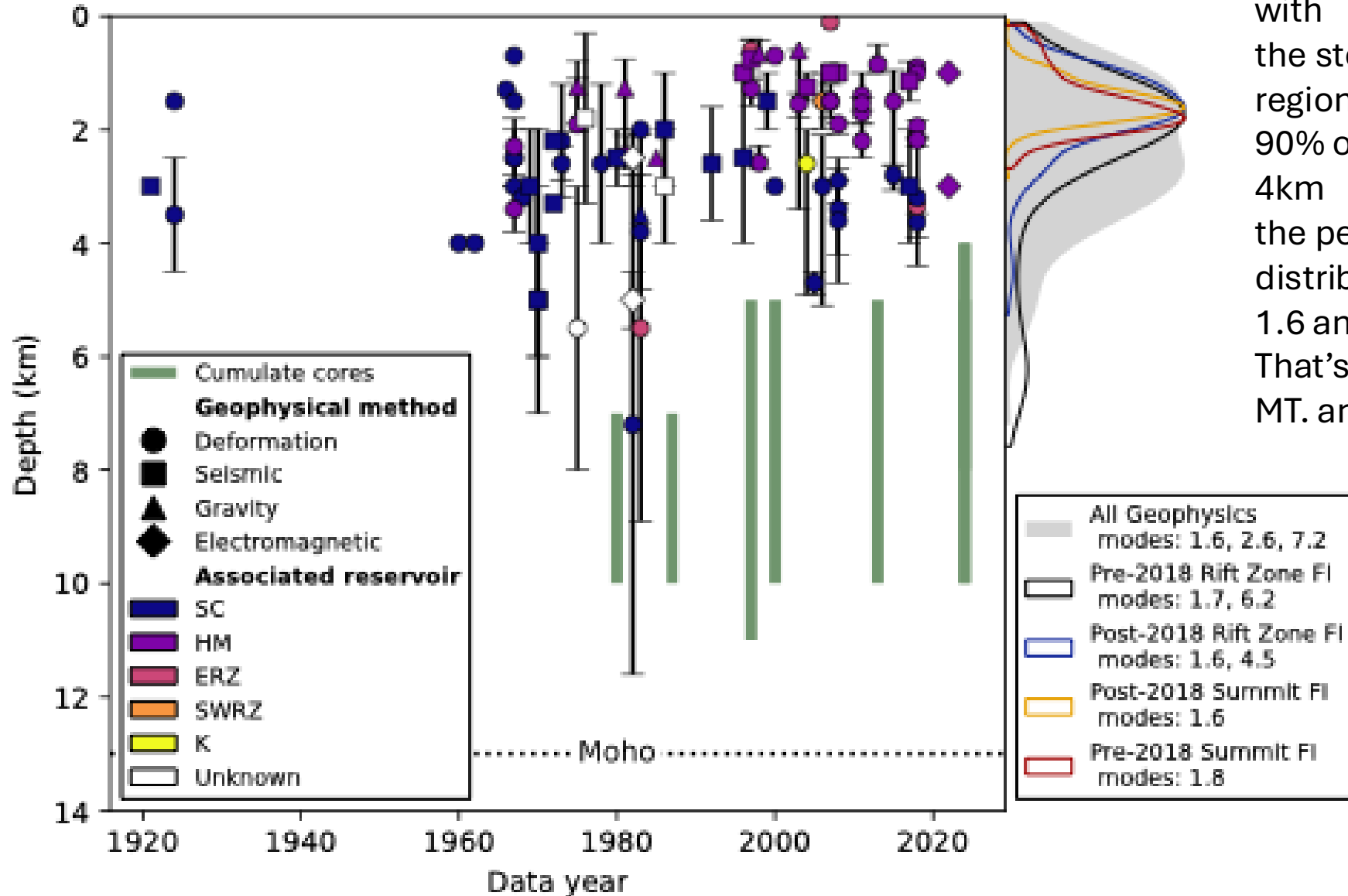


1500 years shallow Kilauea project

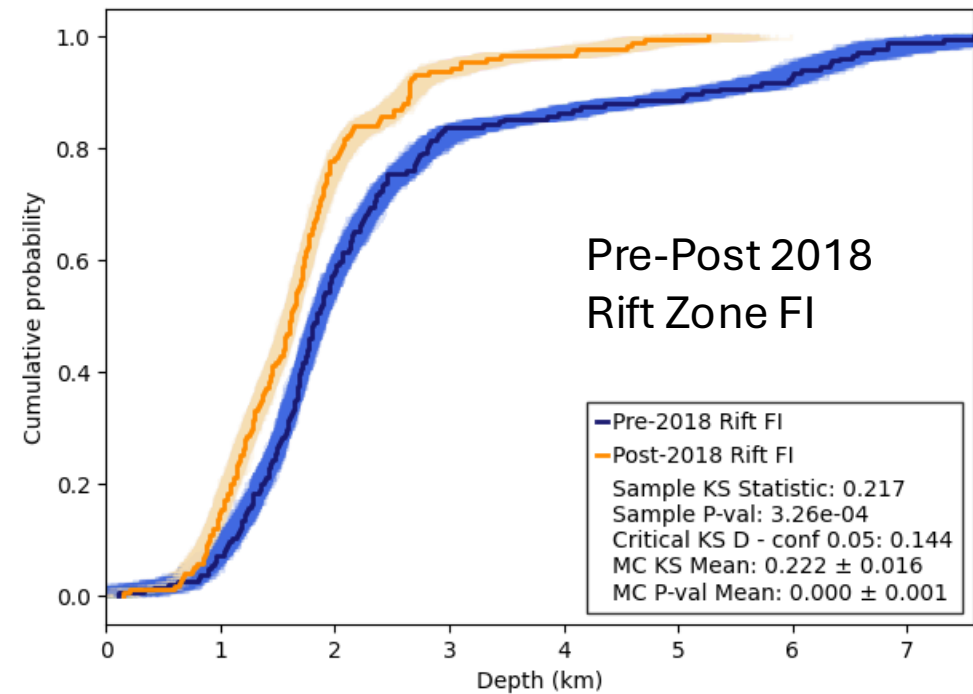
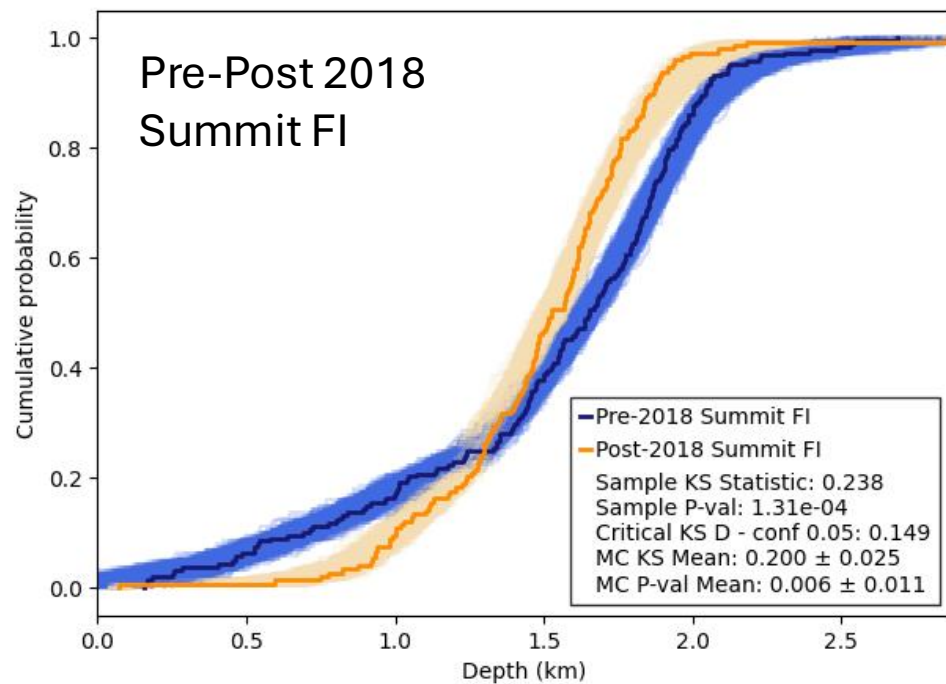
Updated Dec3 2025

Charlotte Devitre

I've left a couple of interesting very recent papers at the end, the rest are compiled in the excel and a separate pptx



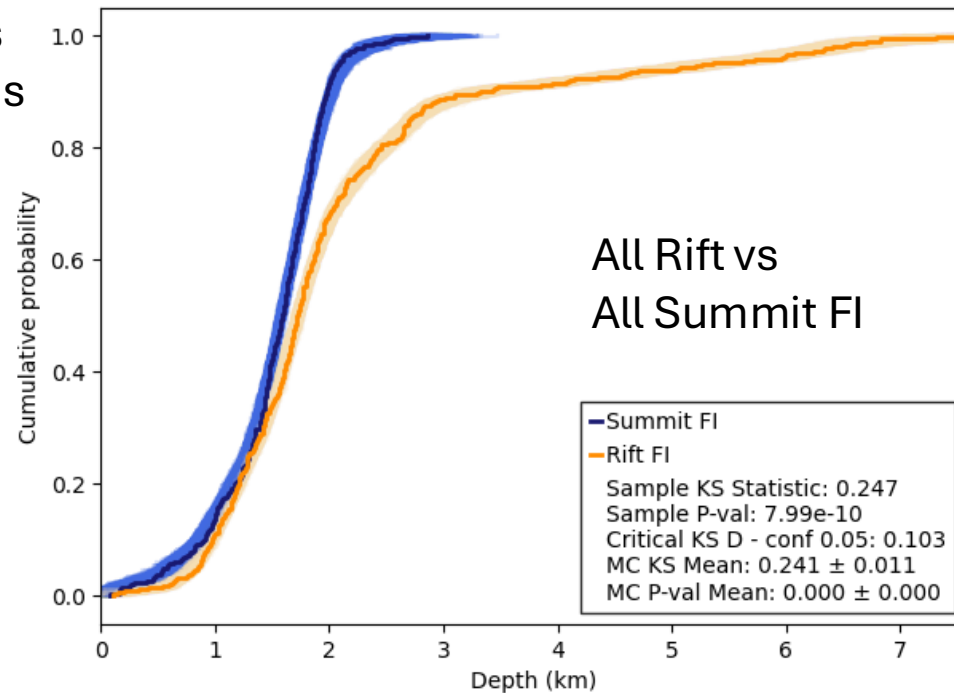
Geophysics is more consistent with the storage in south caldera region <4km and not up to 5. 90% of the geophysics is above 4km the peaks in the geophysics distributions would suggest 1.6 and 2.6 centroids. That's consistent with the recent MT. and with the FI data

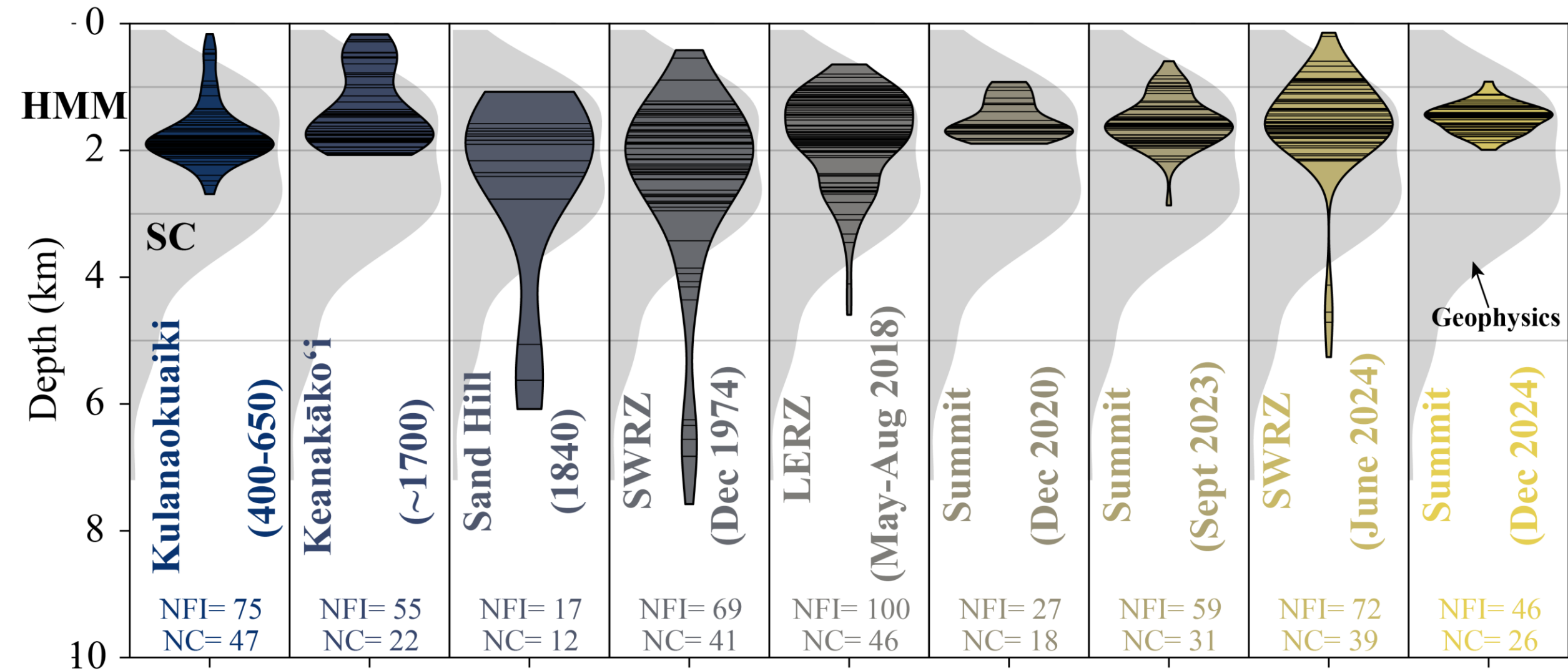


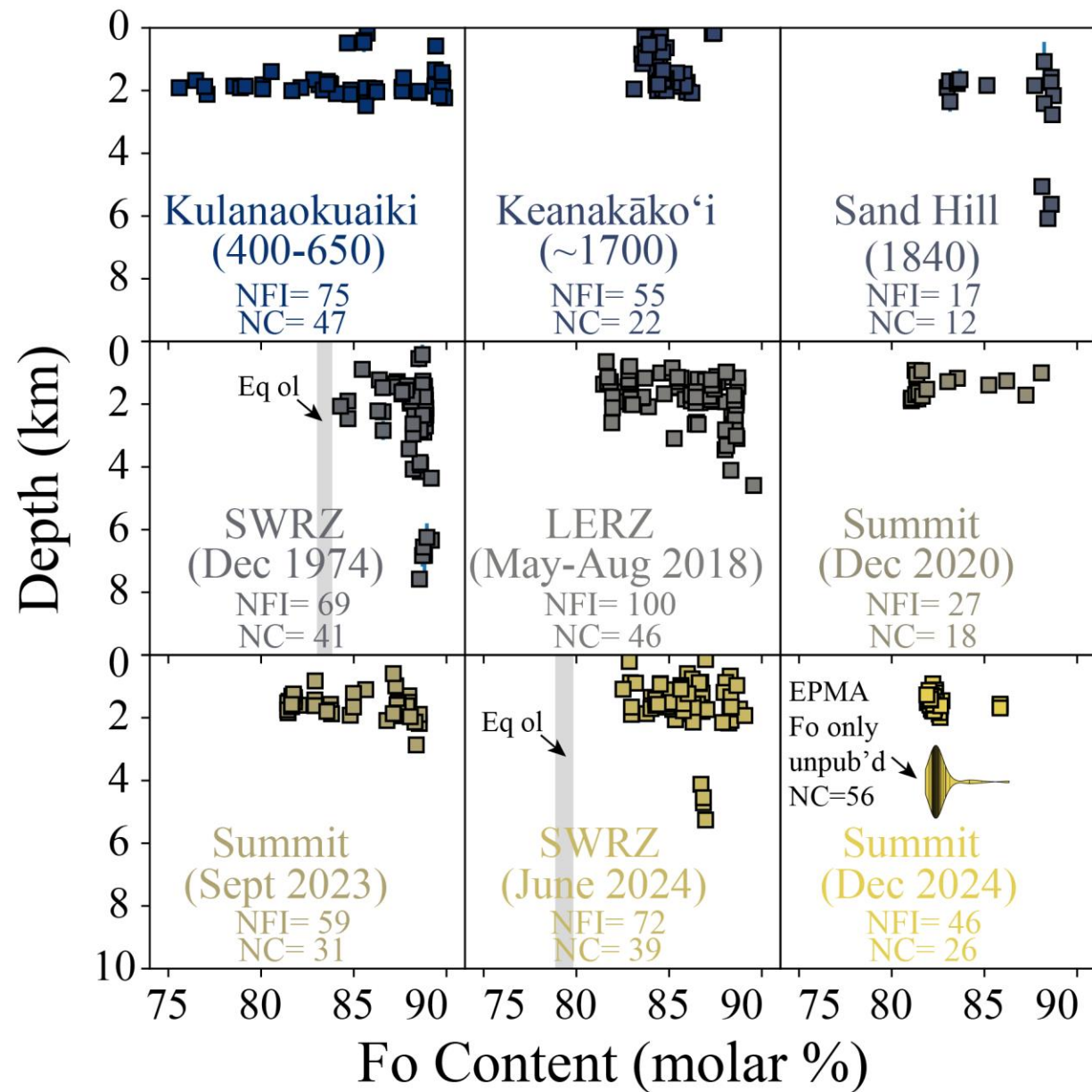
Both Rift and Summit FI populations are shallower for post 2018 eruptions
 And in general Rift is significantly deeper than Summit eruptions

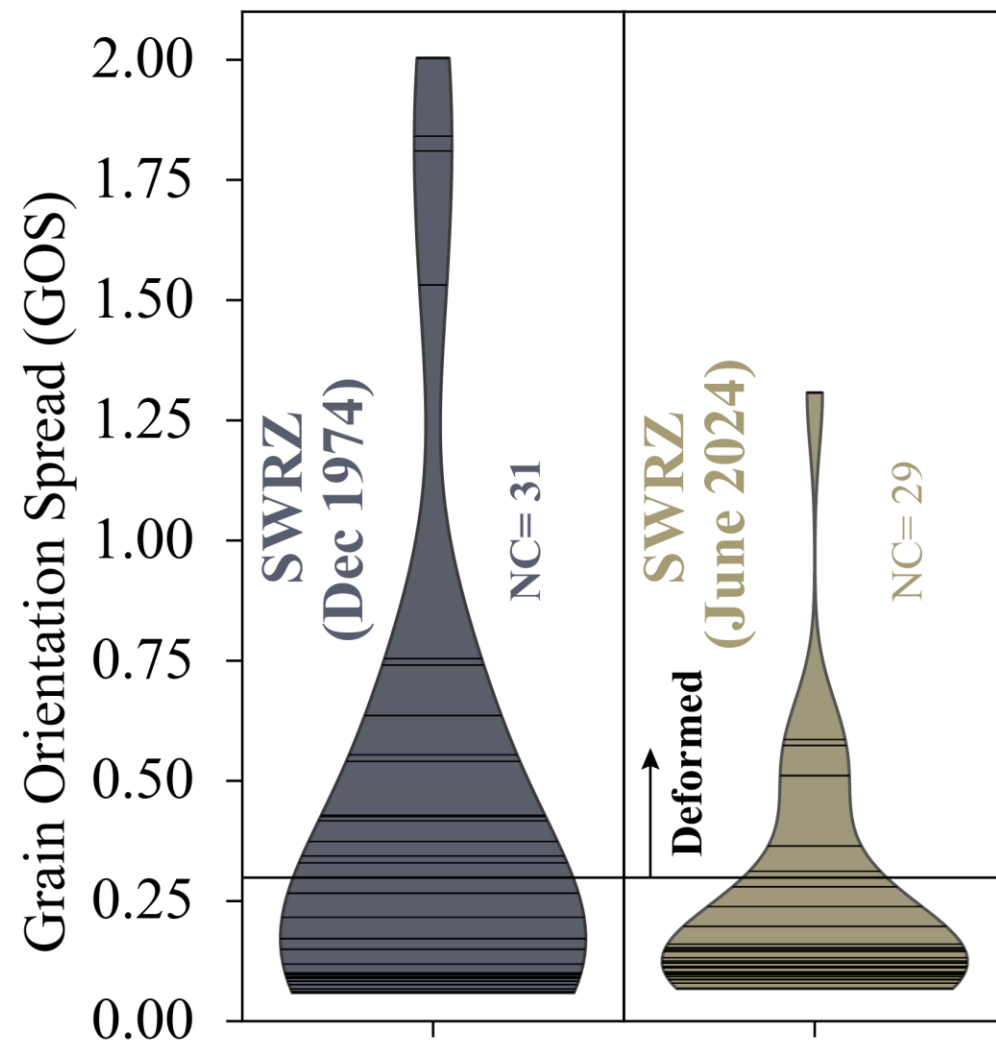
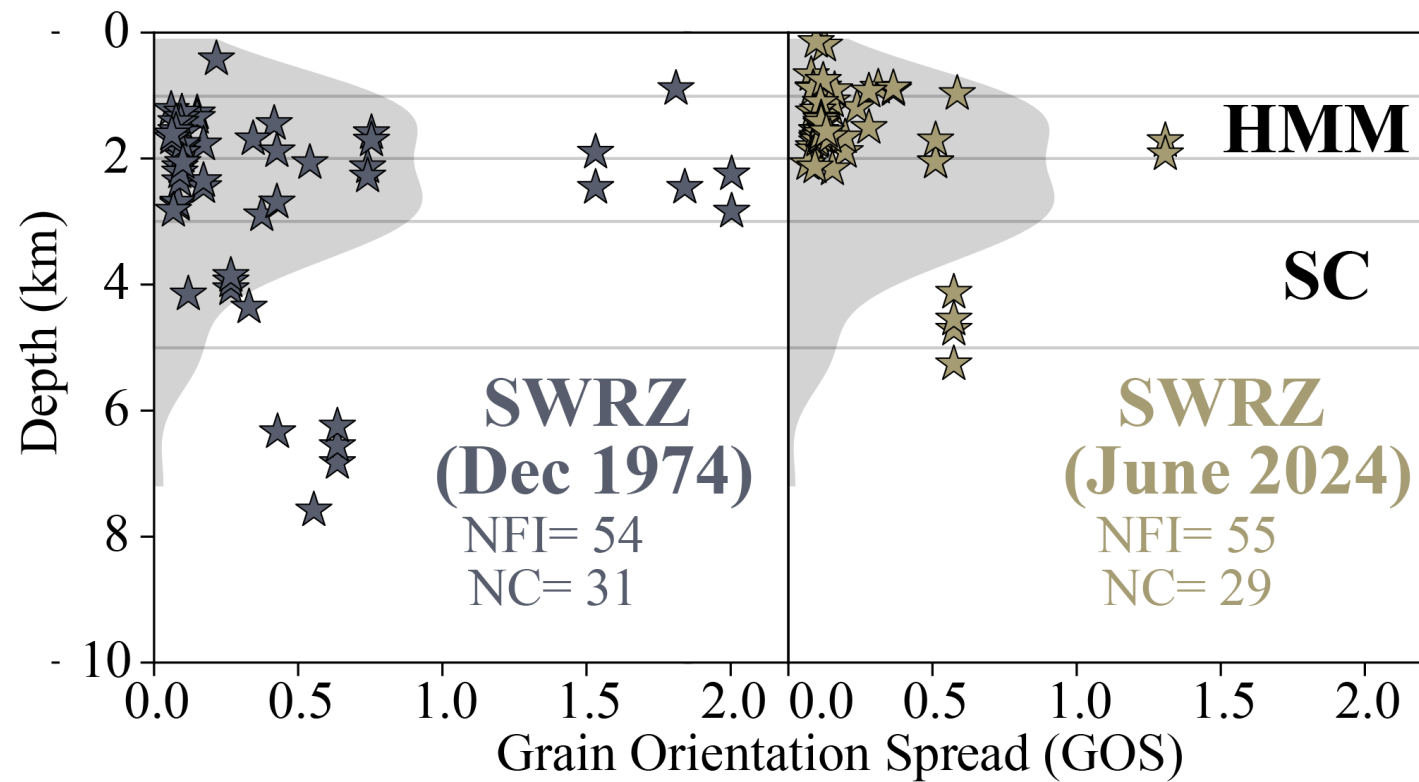
Shallowing over time is supported by this.

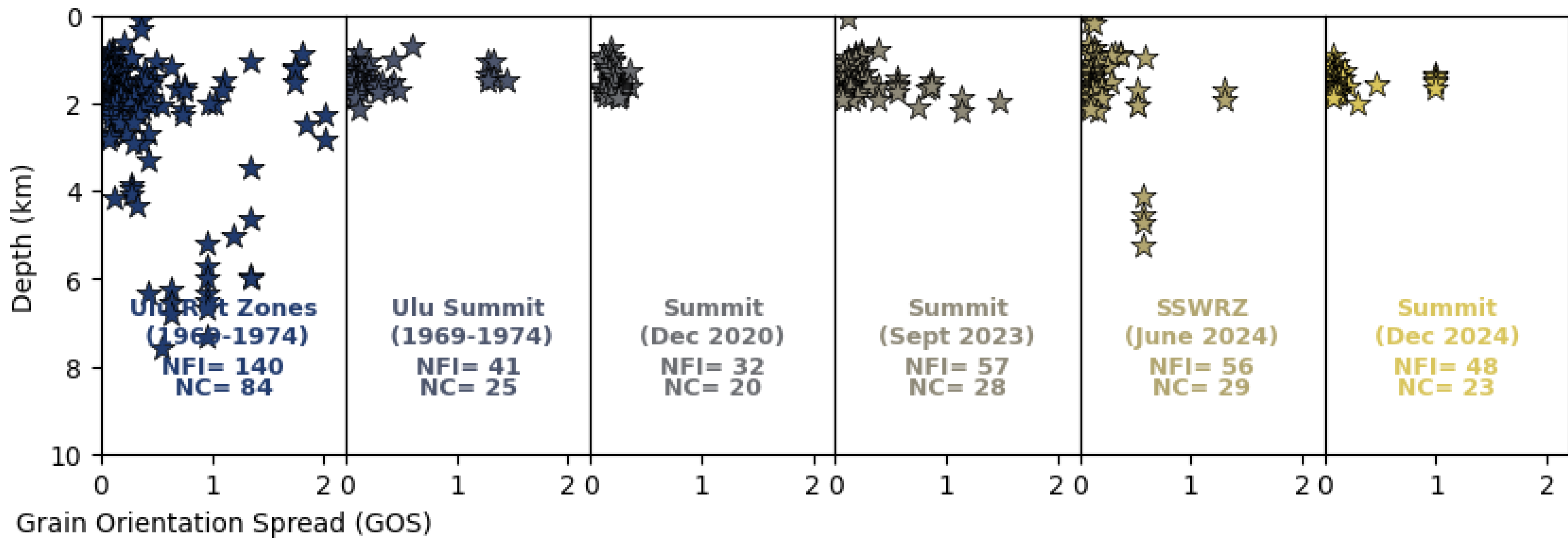
Hard to say if something to do with 2018 or if it's since 2000's but at the very least since 2018 things are shallower

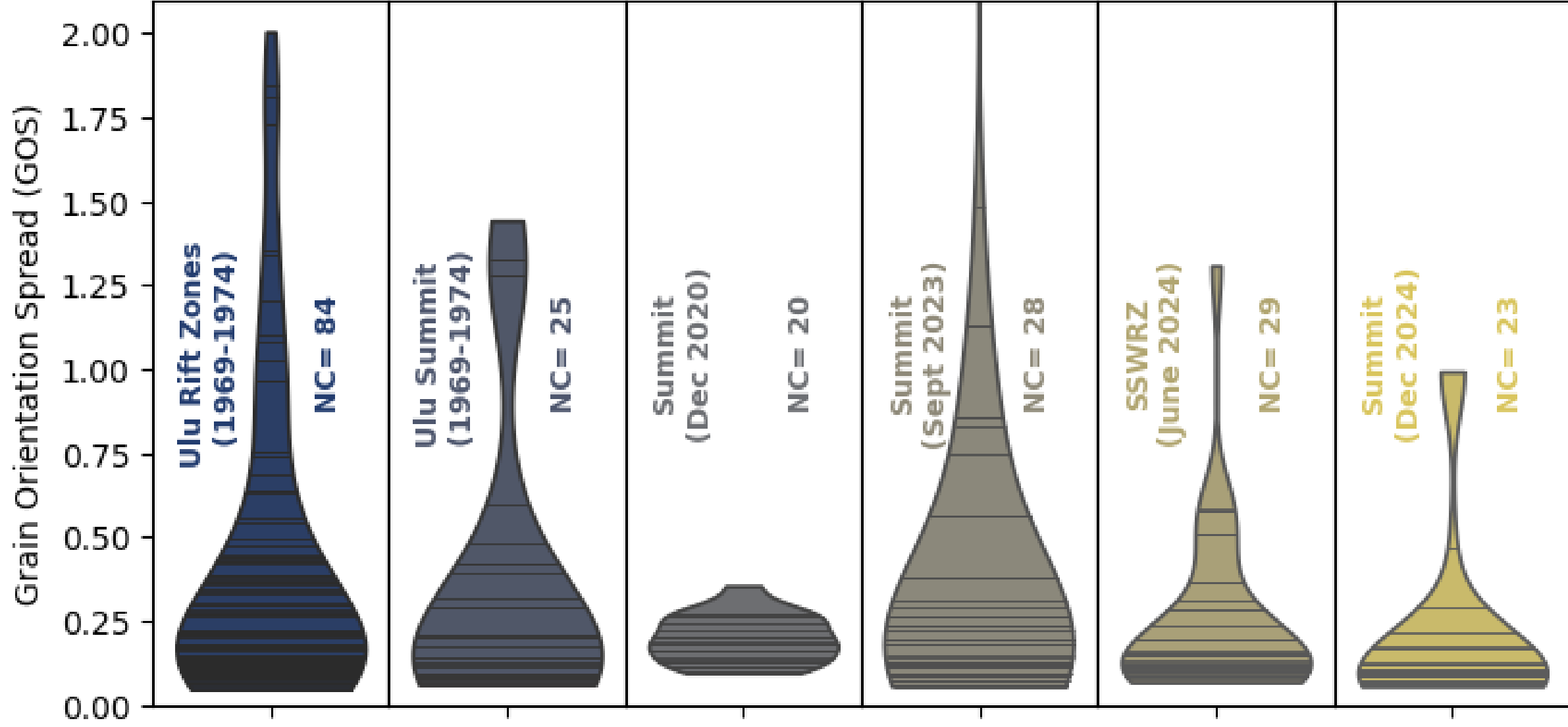


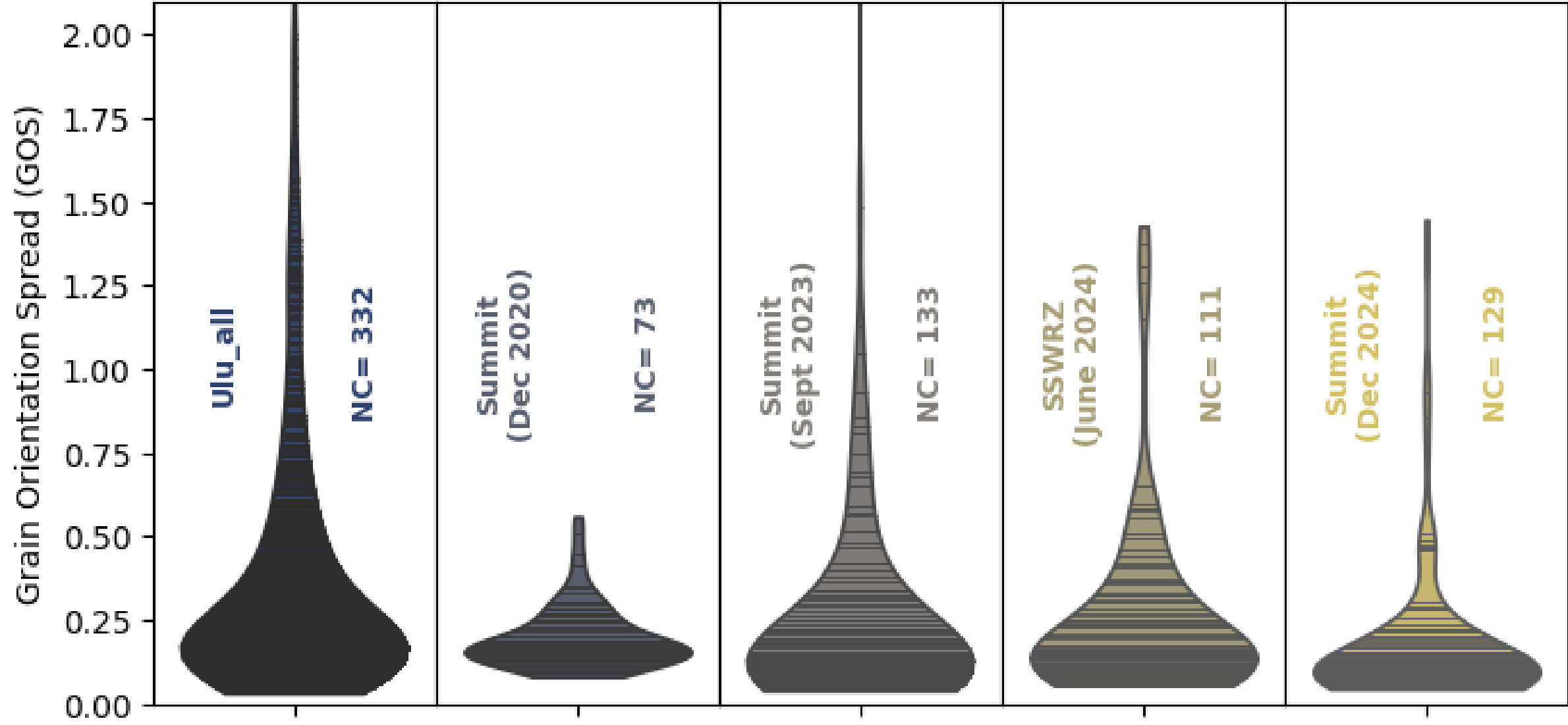


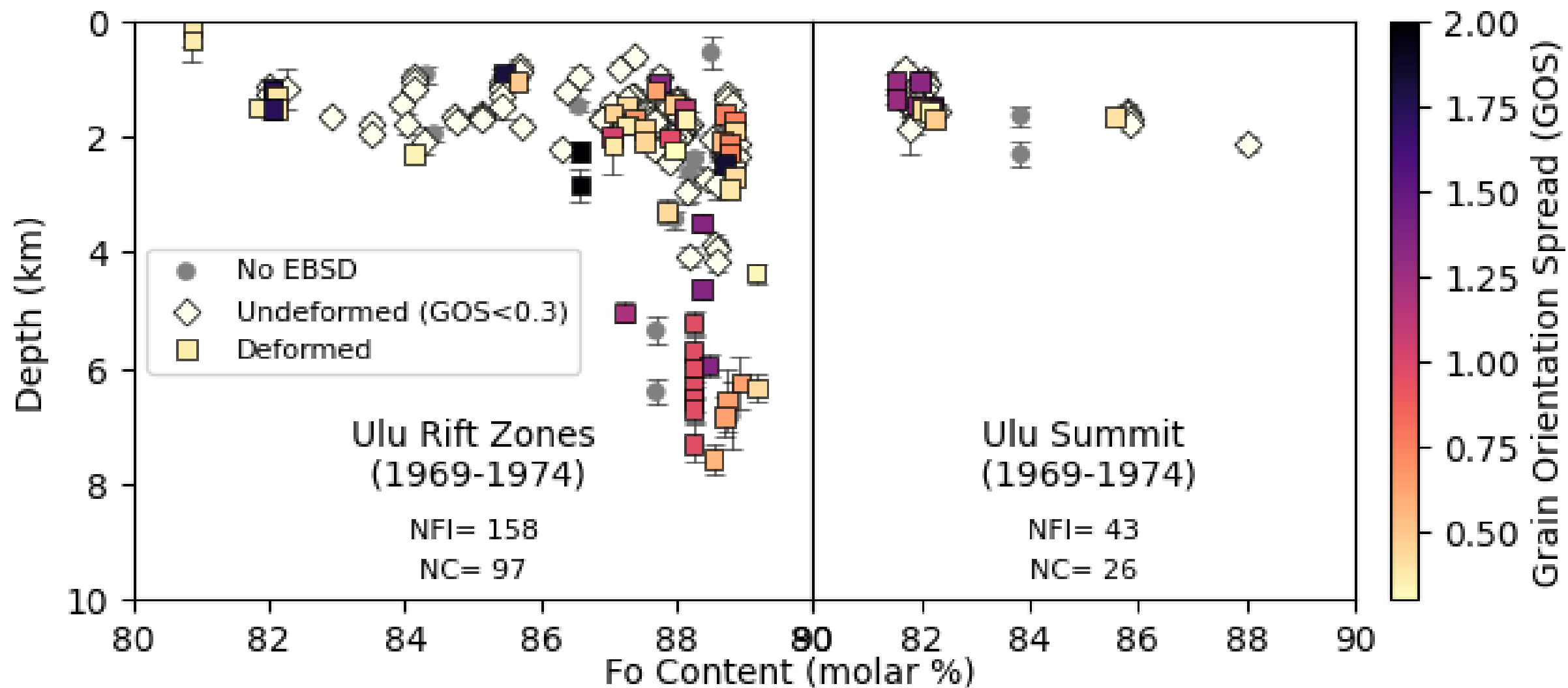


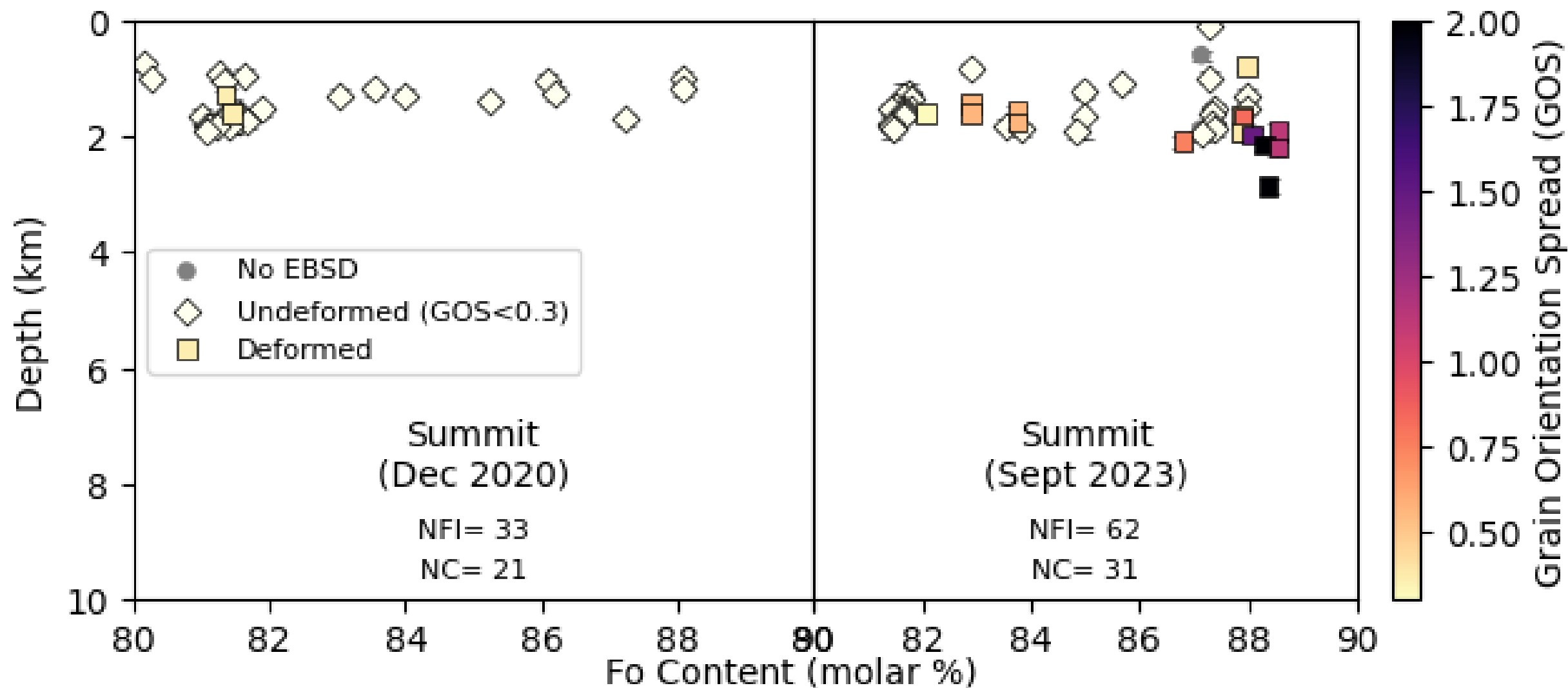


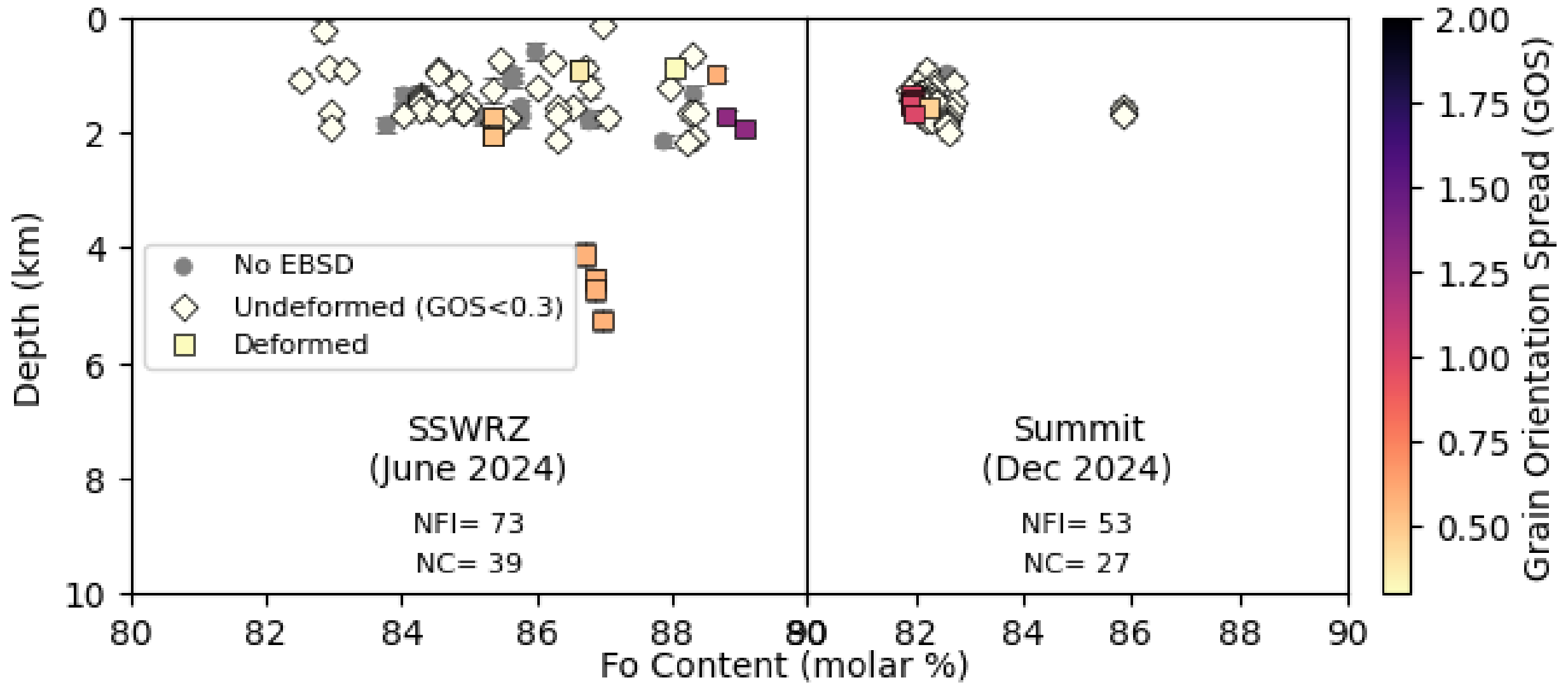


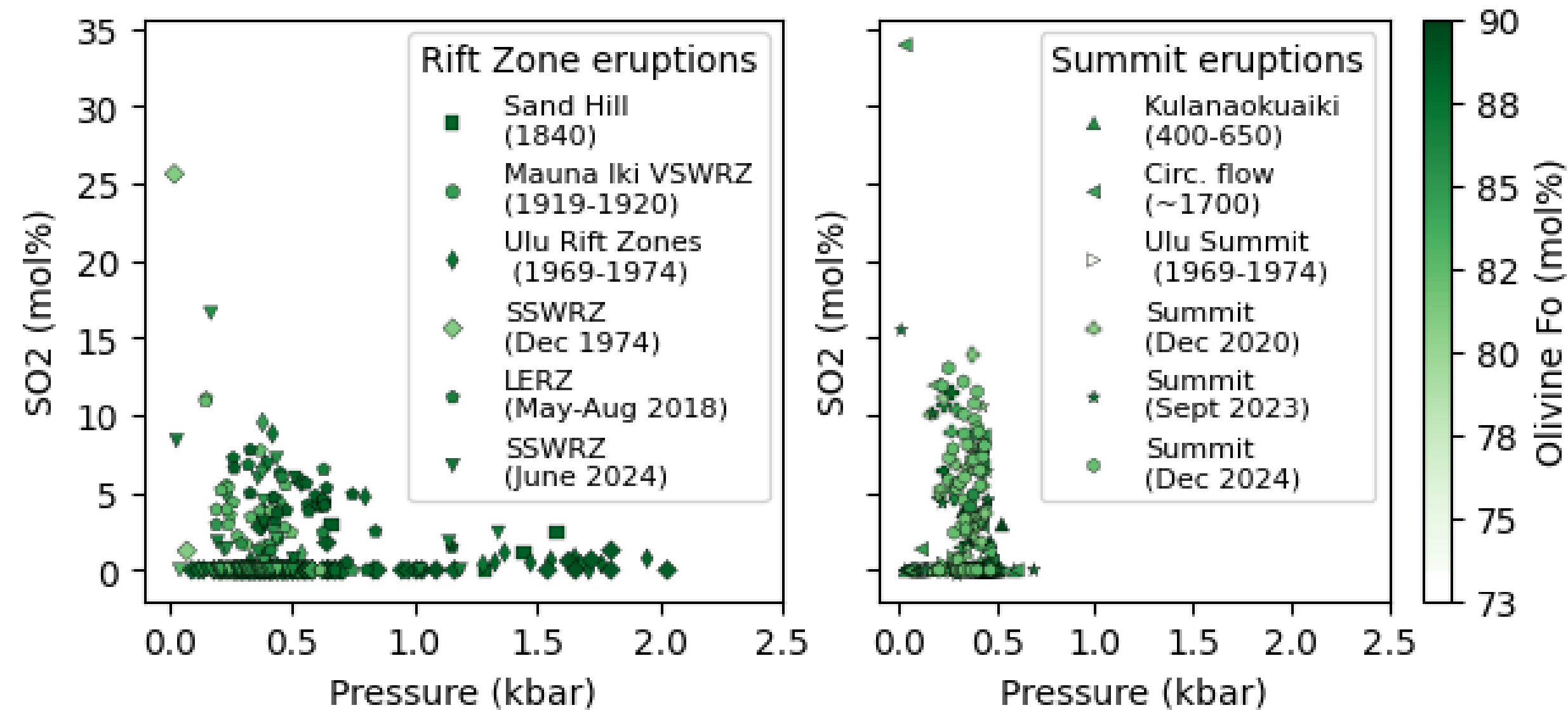


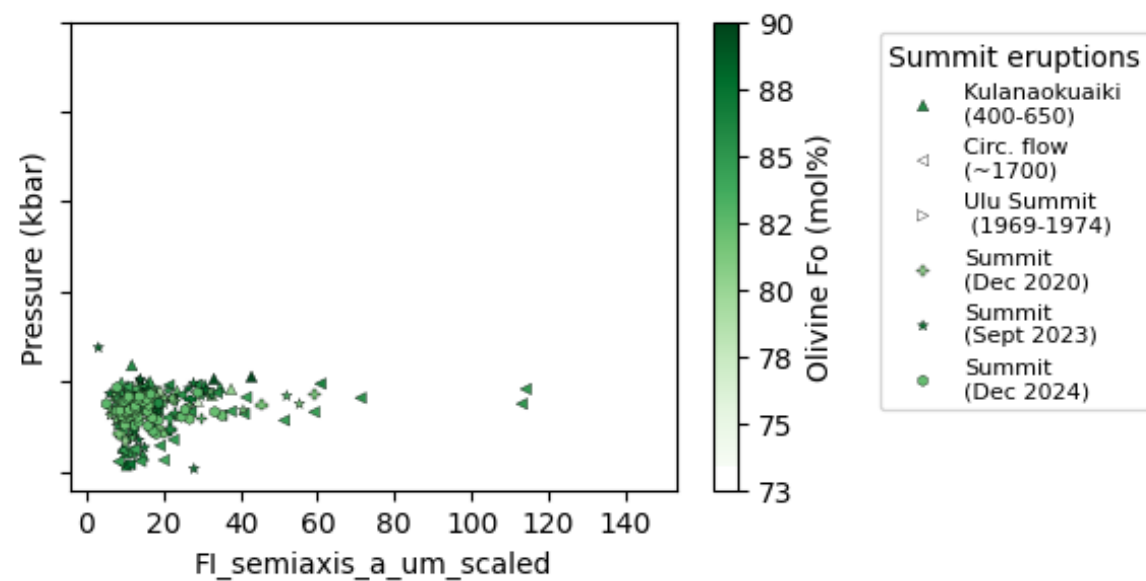
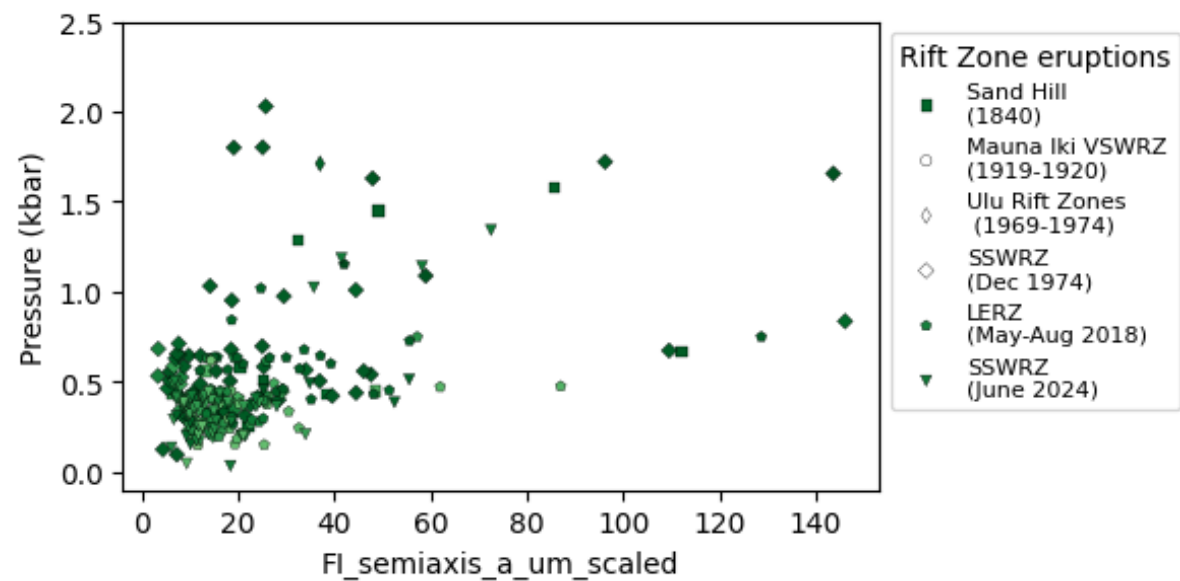












Literature (2000 and on)

Wang 2021

Simultaneous inflation and deflation of HMM and SC

Reservoir associated	Depth (km)	-km uncertainty	+km uncertainty
HM	2.18	0.06	0.01
SC	3.63	0.15	0.28
ERZ	3.35	1.05	1.05

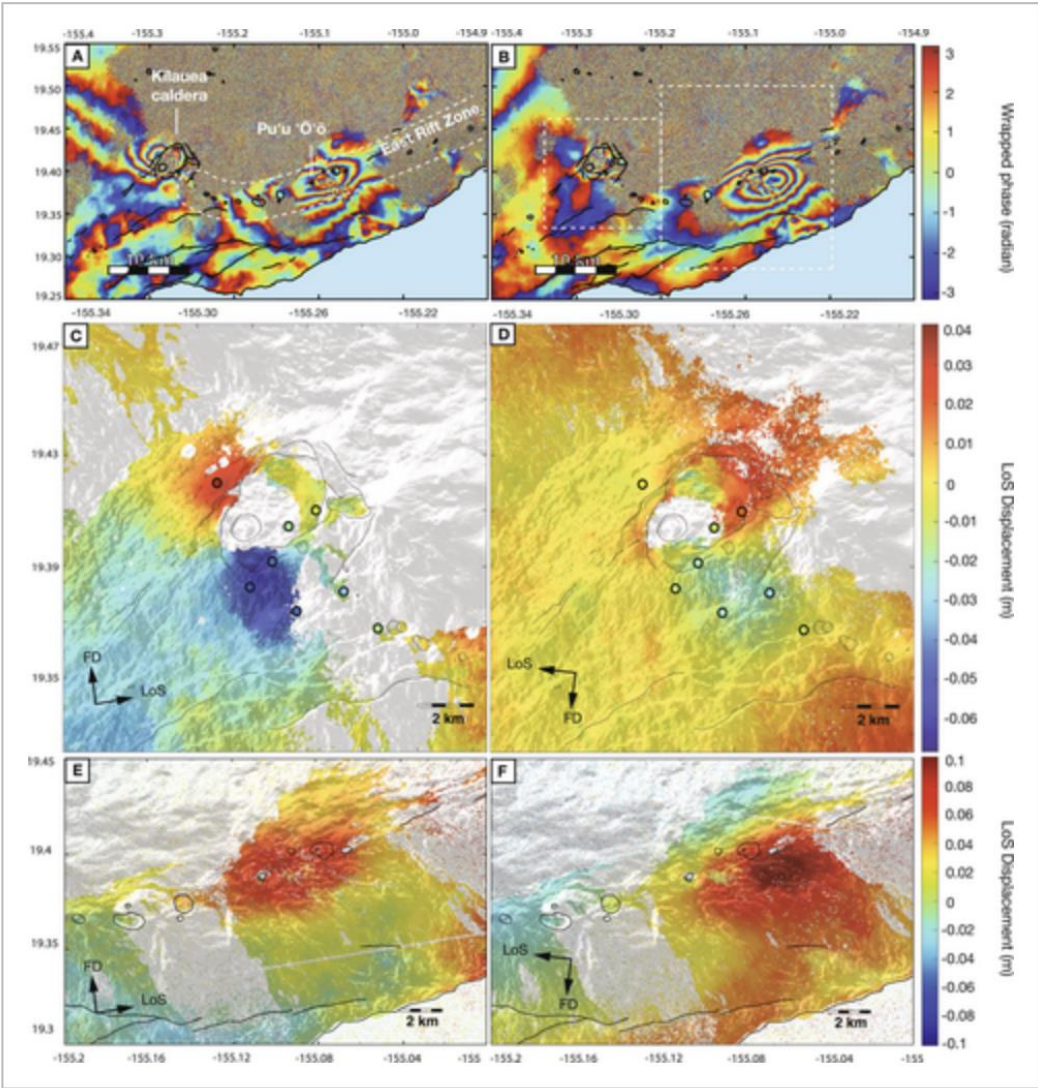


Figure 1 [Open in figure viewer](#) [PowerPoint](#)

Post-collapse simultaneous inflation and deflation at Kilauea summit and inflation in the East Rift Zone. (a), (b): ascending (November 22, 2018–May 27, 2019) and descending (November 13, 2018–May 30, 2019) wrapped interferograms of the summit region and the East Rift Zone. Each color cycle (red-yellow-blue)

Literature (post Cervelli and Milkus)

Hoversten 2022

2002-2003 MT data inverted in 2022; 3D MT, suggests summit reservoir system 1-3 km below summit, centered ~2.

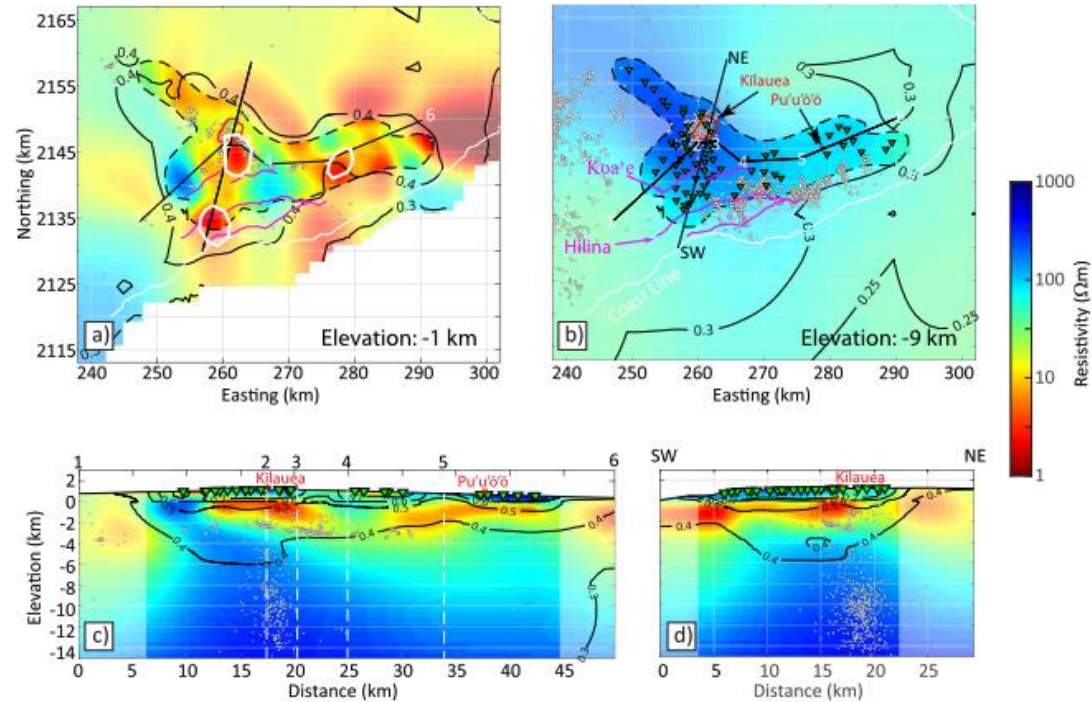


Figure 11. Four slices through the 3D magnetotelluric (MT) inversion model. (a) Is the elevation slice at -1 km. The southwest to northeast and center rift zone traverses are shown as black lines, the Koa'e and Hilina fault systems are marked with magenta lines. Three white polygons in (a) mark volumes used for conductivity averaging. (b) Is the elevation slice at -9 km. The MT sites are shown as green triangles. (c) Is a depth-section traverse along the center of a polygon defining the interpreted rift zones at -1 km elevation from Ryan (1988). (d) Is southwest to northeast depth-section through Kilauea caldera. The black rimmed white dots are earthquake hypocenters from Lin, Shearer, et al. (2014) within the time range 2001 thru 2004 that lie within 2 km perpendicular distance from the section lines (c and d) and between the surface and -2 km elevation for the -1 km elevation slice and between -7 and -11 km elevation for the 9 km elevation slice. The black contour lines are constant values of the approximate resolution matrix. Areas below resolution value of 0.3 or outside of the polygon around the MT sites are shaded.

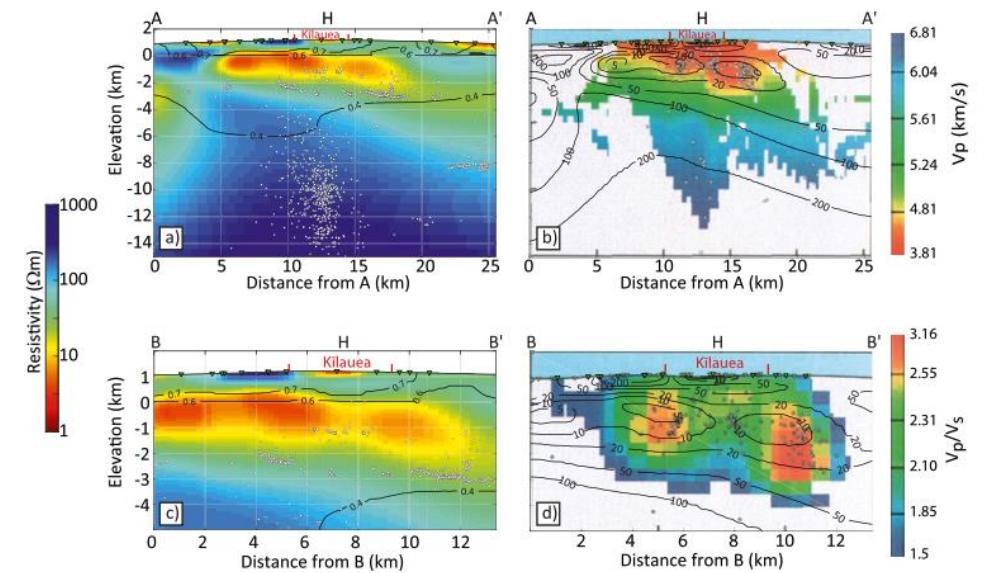


Figure 12. Comparison between the 3D magnetotelluric inversion and published V_p and V_p/V_s models from Dawson et al. (1999). (a) is the 3D resistivity inversion on transect A-H-A' of Dawson et al. (1999) with contours of approximate resolution matrix overlaid. (b) Is the V_p digitized from Figure 3a of Dawson et al. (1999) with contours of resistivity from (a) overlaid. (c) Is the 3D resistivity inversion from transect B-H-B' of Dawson et al. (1999) with contours of approximate resolution matrix overlaid. (d) Is the V_p/V_s digitized from Figure 3b of Dawson et al. (1999) with contours of resistivity from (c) overlaid. Only resolution contours of 0.7 or less are displayed on (a) and (c), higher values cluster near the surface. See Figure 11 for definition of symbols.

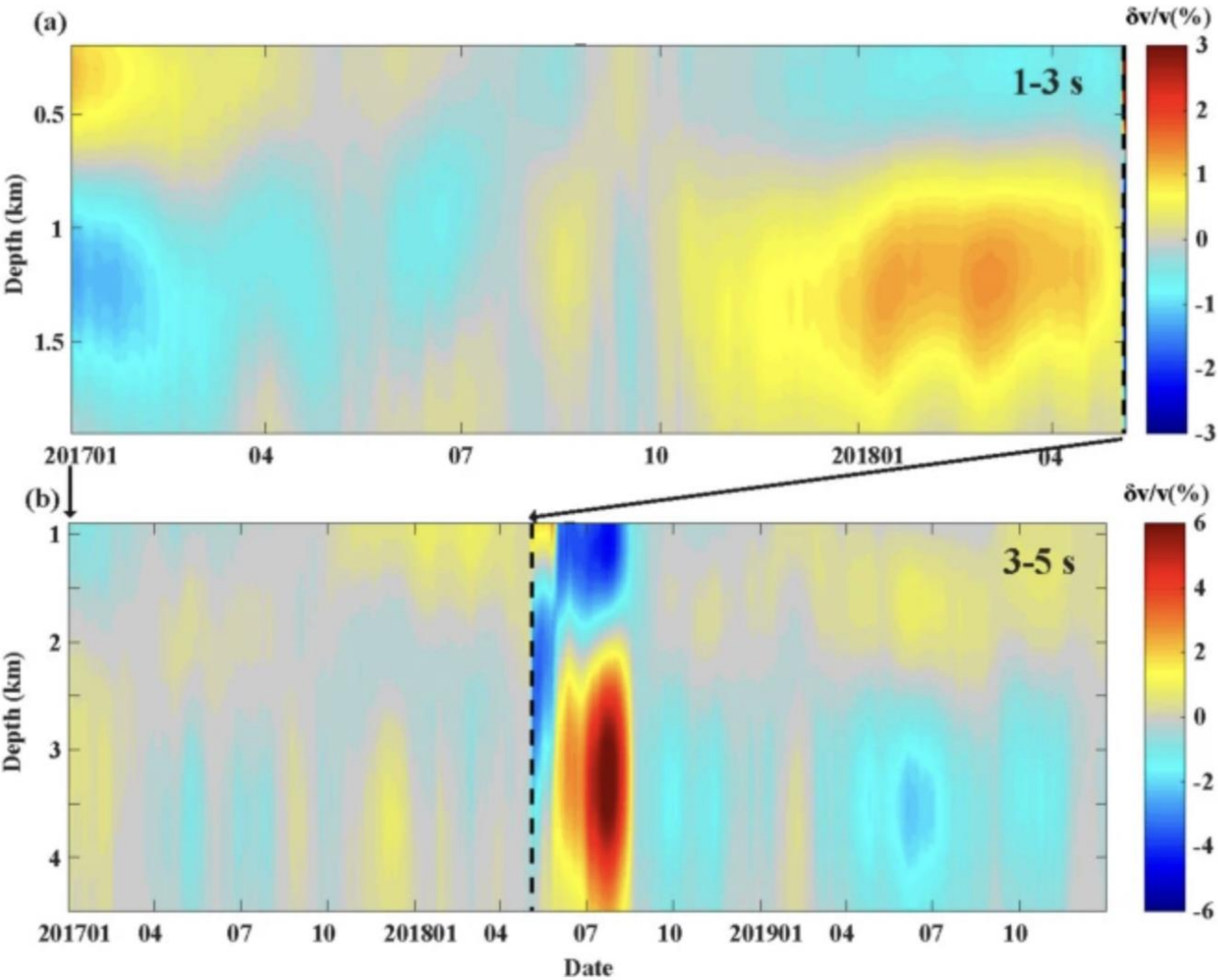
Literature (2000 and on)

Zhou 2025

1-1.5 pressurization prior to onset of 2018, then in July pressurization deeper at 2-4.

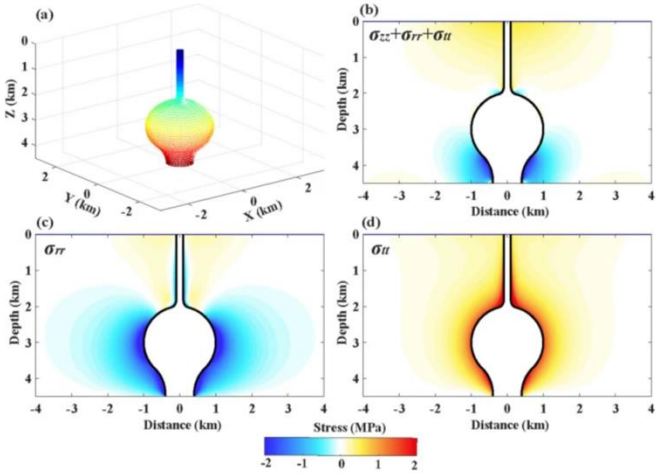
They suggest that first fed from HMMR, then influx from SCR, which couldn't go to HMMR...maybe separate connections like Anderson 2020 suggests?

Fig. 9



Depth-dependent relative shear wave changes obtained by inverting frequency-dependent average $\delta v/v$ for the periods of 1-3 s and 3-5 s. a and b The inversion results based on the data from Figs. 5g and 6g, respectively. Vertical black dashed lines mark May 4, 2018

Fig. 10



Stress produced by upwelling magma in an elastic half space. a The 3D reservoir model, showing the magma reservoir and its conduits. b Normal stress. c Radial stress. d Tangential stress. The black solid line outlines the magma reservoir and its conduits, with the reservoir having a radius of 1000 m, the upper conduit a radius of 100 m, and the lower conduit a radius of 400 m. Red indicates tension, and blue indicates compression